Gmail

Sonu stories <dhivyasonu92@gmail.com>

Assignment 2

Sonu stories <dhivyasonu92@gmail.com>
To: Giridhar M <mgiricse@gmail.com>

---------- Forwarded message ---------
From: **Sonu stories** <dhivyasonu92@gmail.com>
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Subject: Assignment 2
To: Sonu stories <dhivyasonu92@gmail.com>

Tasks:

1. Handling Missing Values:

Check for missing values in the 'Price' column. How would you handle products with missing price information?

There were 3 products with missing prices:

Sony Headphones
Coleman Camping Tent
Ray-Ban Sunglasses

To fill these missing values, I used a hierarchical imputation approach:

Step 1: Use the Average Price of the Same Product

For each product with a missing price, I first checked whether the dataset contained other records with the same product name and valid prices.

For example:

If a product named "Laptop" had a missing price, I calculated the average price of all other "Laptop" records and used that value.

This preserves product-specific pricing patterns.

Step 2: Use the Category Average if No Product Match Exists

Some products, such as Camping Tent, appeared only once in the dataset and had no valid prices available for the same product.

In such cases, calculating the product average resulted in a #DIV/0! error because there were no prices to average.

To handle this, I used the average price of products within the same category.

For example:

Camping Tent belongs to the Outdoor category.
Since no other Camping Tent prices existed, I used the average price of products in the Outdoor category.
Step 3: Keep Existing Prices Unchanged

For products that already had a valid price, I retained the original value and did not modify it.

Formla Used : =IF(D2="",IFERROR(AVERAGEIF($B$2:$B$36,B2,$D$2:$D$36),AVERAGEIF($F$2:$F$36,F2,$D$2:$D$36)),D2)

2.Handling Missing Categories

There were several records with missing category values. Since Category is a categorical field, it cannot be filled using numerical methods such as averages or medians. Therefore, a lookup-based im

Step 1: Use Product Name to Infer Category

First, I checked whether the same product appeared elsewhere in the dataset with a valid category.

For example:

Product Name	Category
Sneakers	Fashion
Sneakers	Blank

Since another Sneakers record already belonged to the Fashion category, I assigned Fashion to the missing record.

The same approach was used for:

Sneakers → Fashion
Coffee Maker → Kitchen
Fitness Tracker → Electronicscs

This is the most reliable method because products typically belong to the same category regardless of brand.

Step 2: Use Brand Name as a Fallback

If no matching product name was found, I searched for other products from the same brand that had a valid category.

For example:

Brand	Product	Category
North Face	Jacket	Outdoor
North Face	Backpack	Blank

Since North Face products are associated with the Outdoor category, the missing category could be inferred from the brand.

This provides a secondary source of information when product-level matching is not possible.

Step 3: Flag Remaining Records for Manual Review

If neither Product Name nor Brand Name provided enough information, I did not assign a category automatically.

Instead, the record was marked as Manual Review.

This prevents introducing incorrect classifications and maintains data quality.

Formla used :`=IF(F2<>"",F2,IFERROR(INDEX(FILTER($F$2:$F$36,$B$2:$B$36=B2,$F$2:$F$36<>""),1),IFERROR(INDEX(FILTER($F$2:$F$36,$C$2:$C$36=C2,$F$2:$F$36<>""),1),"Manual Review`

2. Correcting Inconsistent Data:

I identified inconsistent text formatting in the Product Name column, where some products were stored in lowercase (e.g., "laptop", "smartphone", "headphones") while others used proper case ("Laptop", "Smartphone", "Headphones"). These inconsistencies can cause duplicate groupings and inaccurate analysis. To standardize the data, I converted all product names to a consistent format using the PROPER() function and removed any unnecessary spaces using TRIM(). This ensures that identical products are treated as a single entity during analysis.

3. Removing Duplicates:

A duplicate row is a record where all column values are exactly the same as another record.

Duplicate Rows Found

In your dataset, the following records appear more than once:

Duplicate 1

Product ID	Product Name	Brand	Price	Quantity	Category
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories

Duplicate 2

Product ID	Product Name	Brand	Price	Quantity	Category
16-APR-ES	headphones	Bose	250	20	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni

Duplicate 3

Product ID	Product Name	Brand	Price	Quantity	Category
17-JUN-IN	Laptop	HP	950	25	Electronics
17-JUN-IN	Laptop	HP	950	25	Electronics

Summary

Metric	Count
Total Records Before	35
Duplicate Rows	3
Records After Removing Duplicates	32

How to Remove Duplicates in Google Sheets
Method 1: Built-in Remove Duplicates
Select the entire dataset.
Click Data → Data cleanup → Remove duplicates.
Check Data has header row.
Click Remove duplicates.

4. Splitting and Merging Data:

Used the Text to Columns (Split Text) feature to split the Product ID into Manufacturing Date and Country Code. Then used the CONCATENATE (CONCAT) function to merge the Brand Name and Product Name columns into a new Product Brand column. Any unnecessary characters were removed during the transformation process.

5. Number Formatting:

Formatted the Price ($) column using the Currency number format to display monetary values consistently. Formatted the Manufacturing Date column using the custom date format DD-MM-YYYY to ensure a standardized date representation across the dataset.

6. Conditional Formatting:

1. Data Bars / Color Scales for the "Price ($)" Column

Applied Conditional Formatting to the Price ($) column using Data Bars (or Color Scales).

This provides a visual representation of product prices, where higher-priced items display longer bars (or darker colors) and lower-priced items display shorter bars (or lighter colors).

This makes it easier to identify high-value and low-value products at a glance.

2. Custom Rule for the "Category" Column

Created a custom conditional formatting rule to highlight all cells where the category is "Electronics".

Used the rule: